

# Numerical Verification of a Curvature-Bound Bounce Cosmology: Effective-Field-Theory Control Pushes the Transition Below the Planck Scale

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## Abstract

Classical general relativity can lead to singularities of divergent curvature and density in black-hole interiors and in a backward extrapolation of the early Universe. This paper takes a minimal phenomenological framework — a coordinate-invariant curvature bound, an effective non-singular bounce at a critical density, and a distinction between an event horizon and the genuinely high-curvature region — and reports what happens when its entries are computed rather than asserted. The location of the curvature bound is among them: it is not assumed to be Planckian but computed, and effective-field-theory control places it six to sixteen orders of magnitude below the Planck curvature, at  $\rho_c^{1/4} \sim 10^{16}\text{--}10^{17}$  GeV. The effective bounce background is shown to coincide *exactly* with a background ansatz for which a stable  $c_T = 1$  beyond-Horndeski construction already exists in the literature, with an effective Friedmann residual of  $2.8 \times 10^{-17}$ . Within that construction a  $w(t)$  transition background is ghost-free, gradient-stable and subluminal throughout, and suppresses shear. Two routes are excluded in the process: the minimal single-field realisation, and the constant-equation-of-state baseline, the latter because scalar subluminality requires  $w \leq 1$  while shear suppression requires  $w > 1$ . Integrating the Mukhanov–Sasaki equation with the  $z''/z$  that follows from the surviving trajectory gives  $n_s \simeq 3.35$ , so the adiabatic mode of the bounce cannot originate the observed fluctuations; an entropic mechanism repairs this at the cost of tuning a tachyonic entropy mass to 0.29%. The coherent-relic envelope is shown *not* to be derivable from the background: the transfer function carries no band-limited feature, and an isotropic background cannot generate angular structure at all. The only dynamical source of a direction is residual shear, which forces a quadrupole with a predicted scale dependence  $g_*(k) \propto k^{-0.70}$ , reducing the observational template from six free parameters to two. The amplitude remains unpredicted, so only an upper limit follows. No observational data was used in this work.

**Keywords:** quantum gravity; Planck scale; curvature invariant; bounce cosmology; beyond-Horndeski; statistical anisotropy; early Universe

## 1 Problem statement

In general relativity, gravity is spacetime curvature rather than a single force strength. The classical interior solution of a black hole and a backward extrapolation of FLRW cosmology both reach, under stated conditions, a singularity of divergent curvature. It is more conservative to read that divergence as a signal that the classical theory is insufficient in the region than as an actual infinity in nature.

One distinction matters throughout. The **event horizon** of a large black hole can be locally weakly curved and is not automatically a quantum-gravity region. The transition candidate considered here is not the horizon but the region where spacetime curvature or matter density approaches the Planck scale.

## 2 Assumptions and scales

### 2.1 Planck-scale benchmark

Combining  $\hbar$ ,  $G$  and  $c$  gives the characteristic length and density

$$\ell_{\text{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad \rho_{\text{P}} = \frac{c^5}{\hbar G^2}. \quad (1)$$

These are dimensional benchmarks at which quantum effects and gravity are expected to become simultaneously non-negligible, not a proof of a unified theory.

### 2.2 Curvature-bound hypothesis

A “maximum gravity” is properly expressed through a coordinate-invariant curvature scalar rather than an observer-dependent acceleration. For the Kretschmann scalar  $\mathcal{K} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$  we assume that a *finite* bound exists,

$$\mathcal{K} \leq \mathcal{K}_{\text{max}}. \quad (2)$$

This is the only curvature assumption made, and no microscopic origin for it is supplied.

*The location of the bound is not assumed.* The Planck length is a natural unit in which to express equation (2), not a claim that the bound sits there. Defining

$$\eta \equiv \mathcal{K}_{\text{max}} \ell_{\text{P}}^4 \quad (3)$$

makes  $\eta$  a quantity to be computed rather than declared. Section 7 shows that requiring effective-field-theory control gives  $\eta \sim 10^{-16}$ – $10^{-6}$ , so the transition occurs six to sixteen orders of magnitude below the Planck curvature, at  $\rho_c^{1/4} \sim 10^{16}$ – $10^{17}$  GeV.

The proposition this section leaves is therefore not that a bound exists at the Planck scale, but that *a finite bound exists and its location is fixed by the self-consistency of the effective description* — a narrower and more testable statement.

### 2.3 String length scale and the high-curvature crossover

In perturbative string theory the fundamental length is  $\ell_s = \sqrt{\alpha'}$ . The low-energy derivative expansion is trustworthy for  $\alpha'|R_{\mu\nu\rho\sigma}| \ll 1$ ; when  $\alpha'|R_{\mu\nu\rho\sigma}| \gtrsim 1$ , higher-curvature corrections and stringy states cannot be neglected. This motivates, but does not establish, the weaker hypothesis used here: that new degrees of freedom near the string scale may produce a finite effective curvature or a non-singular transition.

For observability, causality is imposed on the coherent interval rather than on the Universe as a whole,

$$L_{\text{coh}}(t) \leq D_{\text{causal}}(t), \quad (4)$$

where  $D_{\text{causal}}$  is the particle or event horizon of the chosen cosmology.

### 2.4 Causality and unitarity

The framework does not assume that causality is lost beyond an event horizon. It assumes instead that a more fundamental quantum causal rule or unitary evolution governs the transition region where the classical description fails. The explicit form of that rule is the central gap of the proposal.

### 3 An effective bounce ansatz

For a spatially flat homogeneous universe with  $c = 1$ , the simplest phenomenological effective equation is

$$H^2 = \frac{8\pi G}{3}\rho \left(1 - \frac{\rho}{\rho_c}\right), \quad \rho_c = \xi\rho_P, \quad (5)$$

with  $\xi$  a dimensionless coefficient that a microscopic theory must supply. Combined with  $\dot{\rho} + 3H(\rho + p) = 0$  this gives

$$\dot{H} = -4\pi G(\rho + p) \left(1 - 2\frac{\rho}{\rho_c}\right), \quad (6)$$

so that for ordinary matter with  $\rho + p > 0$  one has  $\dot{H} > 0$  at  $\rho = \rho_c$ , permitting a non-singular transition from  $H < 0$  to  $H > 0$ .

Equation (5) is an *illustrative ansatz* resembling effective forms widely studied in bouncing cosmology. Writing it down does not derive loop quantum gravity, string theory, or any other specific proposal. Section 7 reports what happens when one asks which microscopic realisations are actually compatible with it.

### 4 Black holes, the Big Bang, and a new expanding region

The minimal narrative of the hypothesis is: gravitational collapse or a black-hole interior evolves to high density; curvature and density approach the Planck bound; a quantum-gravity transition replaces the classical singularity; a bounce occurs; an expanding spacetime region follows.

If that final region is causally disconnected from the parent spacetime, an observer inside it could see a hot dense early universe resembling a Big Bang. This is *not* the claim that the external Hawking evaporation of one black hole is observed as our Big Bang; the externally visible evaporation and the hypothesised interior bounce are distinct problems. Nor does the framework claim that a black hole promptly becomes a white hole in the same external universe: equation (5) is a homogeneous ansatz and cannot simply be inserted into a black-hole interior. A separate spherically symmetric or rotating solution, a transition timescale, boundary conditions on energy and information, and an explicit causal structure would all be required.

Mass should likewise be described as a conversion between hot field and radiation energy and particle masses, not as creation from nothing. Which species are produced, and whether any stable electromagnetically inert remnant could constitute dark matter, requires a separate quantum-field-theoretic calculation.

### 5 Relation to other proposals

Conformal cyclic cosmology connects the far future of one aeon to the Big Bang of the next by a conformal rescaling. The model here does not assume a conformal matching; it focuses on a dynamical rebound at maximum curvature or critical density.

Ekpyrotic and cyclic models are closer, in that they treat a pre-Big-Bang contracting phase and a transition. This work does not assume brane collisions or a specific higher-dimensional theory. At this stage it is therefore not a replacement for those models but a statement of the singularity-avoidance and transition rules they all require.

### 6 Falsifiability and observational program

To become a scientific model, the framework must make at least one quantitative prediction differing from standard inflationary cosmology.

1. **Primordial CMB fluctuations.** *Partly supplied* (§7). The adiabatic tilt  $n_s \simeq 3.35$  is an exclusion; the quadrupolar modulation  $g_*(k) \propto k^{-0.70}$  is a signal whose *shape* is predicted. Its amplitude is not.
2. **Primordial gravitational waves.** *Not supplied.* The surviving construction has  $c_T^2 = 1$  by design and is therefore compatible with GW170817, but that is a constraint the construction was built to satisfy, not a prediction.
3. **Nucleosynthesis and structure formation.** A reheating history compatible with light-element abundances. Not addressed here.
4. **Black-hole observations.** A mass remnant, ringdown deviation, or echo signature cannot be claimed without an explicit matching rule, which §4 does not supply.

## 6.1 Indirect JWST route

JWST cannot observe Planck-scale curvature directly. It can constrain early black-hole seeds and their hosts — for instance through a uniform census of  $z \gtrsim 4$  Little Red Dot candidates in public NIRCам and NIRSpect data, jointly fitting spectroscopic redshift, line width, host stellar mass and lensing selection. Such a program compares astrophysical seed channels; by itself it cannot establish a bounce, a wormhole, or a curvature bound.

## 7 Numerical verification

This section reports only those items of §6 that were actually computed. No observational data was used.

### 7.1 The background lies inside a known stable family

In units  $\alpha = 1$ ,  $a_B = 1$ , the exact constant- $w$  solution of equation (5) is

$$H(t) = \frac{t}{p(1+t^2)}, \quad a(t) = (1+t^2)^{1/(2p)}, \quad p = \frac{3}{2}(1+w), \quad (7)$$

which is *exactly* the background ansatz of Ye and Piao [1] at constant  $p$  and unit lapse — the family for which they give an explicit  $c_T = 1$  beyond-Horndeski construction. The residual of equation (5) on this solution is  $2.8 \times 10^{-17}$ : an identity, not a fit. The ansatz of §3 is therefore not arbitrary phenomenology.

### 7.2 Stability

Two candidates are excluded. A minimally coupled single field  $P(X) = KX + LX^2 - V$  yields no point, out of  $10^6$  sampled, that violates the null energy condition while remaining ghost-free and gradient-stable — consistent with the Horndeski no-go theorem [2]. The constant- $w$  baseline fails for a subtler reason: scalar subluminality requires

$$c_s^2(+\infty) = p/3 \leq 1 \iff w \leq 1, \quad (8)$$

whereas suppressing anisotropy through contraction requires

$$\Delta \ln(\sigma^2/\rho) = (3/p - 1) \ln(1 + \eta_i^2) < 0 \iff w > 1. \quad (9)$$

The two conditions do not overlap. Both boundaries are reproduced to four digits by an exhaustive parameter scan.

A  $w(t)$  transition background ( $w_i \rightarrow 4.33$ ,  $w_f = 0.6$ ) passes every condition, as summarised in Table 1.

| Condition          | Value                               | Verdict            |
|--------------------|-------------------------------------|--------------------|
| No ghost (scalar)  | $\min Q_s = 3.000$                  | pass               |
| No ghost (tensor)  | $\min Q_T = 0.500$                  | pass               |
| Gradient stability | $\min c_s^2 = 0.114$                | pass               |
| Subluminality      | $\max c_s^2 = 0.849$                | pass               |
| Tensor speed       | $c_T^2 = 1$                         | by construction    |
| Shear suppression  | $\Delta \ln(\sigma^2/\rho) = -6.56$ | pass               |
| EFT control        | $\Lambda$ assumed, not derived      | <b>not counted</b> |

Table 1: Stability of the surviving  $w(t)$  transition realisation. The final row is the one entry that passes only because a cutoff was supplied by hand; it is therefore not counted as passed.

### 7.3 Primordial spectrum: the adiabatic mode is excluded

Integrating  $v_k'' + (c_s^2 k^2 - z''/z)v_k = 0$  with the  $z''/z$  that follows from the surviving coefficient trajectory gives  $n_s \simeq 3.35$ , incomparable with the Planck value  $n_s = 0.9649 \pm 0.0042$  [3]. This is a known property of ekpyrotic contraction rather than a surprise. The pipeline was validated against the analytic contracting-phase result

$$n_s - 1 = 3 - 2 \left| \beta - \frac{1}{2} \right|, \quad \beta = \frac{1}{p_i - 1}, \quad (10)$$

which it reproduces to within 0.07 over a factor five in  $p_i$ .

The adiabatic mode of the bounce therefore cannot originate the observed fluctuations. The standard remedy — an entropic mechanism — was tested on the same background. A tachyonic entropy mass  $m_{\text{eff}}^2 = -\lambda H^2$  gives  $n_s = 0.9925$  at  $\lambda = 104$  and  $n_s = 0.9572$  at  $\lambda = 106.6$ . Since  $dn_s/d\lambda = -\beta^2/\nu$ , matching Planck within its error bar fixes  $\lambda$  to 0.29%. That tuning is the price of the repair, and no mechanism in this framework explains it.

### 7.4 The coherent relic: shape predicted, amplitude not

A classical relic obeys the same linear mode equation as the fluctuations, so its envelope is the initial profile multiplied by the transfer function, and only the transfer function is background dynamics. The measured transfer function carries *no* band-limited feature: its cutoff lies two orders of magnitude below the comoving scale of the bounce, so every observable mode is far longer than the transition and carries no record of having crossed it. Splitting the transition into two timescales — as a three-level synchronisation hierarchy would imply — changes nothing, because producing a peak requires modes to exit, re-enter and exit again, and  $|aH|$  remains monotonic in topology.

The angular structure is a stronger statement. On an isotropic FLRW background the mode equation depends on  $|k|$  alone, so no direction dependence can be generated at all. Within this framework the only dynamical source of a direction is residual shear, and it *forces*

$$P(k, \hat{\mathbf{k}}) = P_{\text{iso}}(k) \left[ 1 + g_*(k) (\hat{\mathbf{k}} \cdot \hat{\mathbf{n}})^2 \right], \quad \frac{d \ln g_*}{d \ln k} = -0.70. \quad (11)$$

The multipole is fixed at  $L = 2$ ; the modulation is multiplicative rather than additive; and the tilt agrees across two shear amplitudes, so the shape is a prediction. The  $\mu = (\hat{\mathbf{k}} \cdot \hat{\mathbf{n}})^2$  dependence was measured rather than assumed, and is linear to within 4%.

The free parameters of the observational template accordingly drop from six to two: an overall size and an axis. The initial shear amplitude is *not* fixed by the theory, so only an upper limit follows and detectability is not predicted. Using the Planck quadrupole constraint  $\Delta g_* = 0.016$  [4], the shear amplitude obeys  $C \lesssim 2.3 \times 10^{-6}$ . That published limit assumes a scale-independent  $g_*$ , so this number is indicative pending a dedicated reanalysis with the predicted tilt.

## 7.5 What a black-hole-interior origin would require

If the reading of §4 is maintained, the level of anisotropy becomes the obstacle. A black-hole interior is Kantowski–Sachs rather than FLRW and starts at  $\sigma^2/\rho \sim O(1)$ . Reaching the regime verified above ( $\sim 10^{-7}$ ) demands  $\Delta \ln(\sigma^2/\rho) \simeq -16$ , which at  $p_i = 8$  corresponds to a contraction roughly two hundred times longer than the one computed here. The stability verdict would also have to be re-established non-perturbatively on a Bianchi I background. The black-hole–Big-Bang reading is therefore neither excluded nor supported; what has changed is that the requirement is now quantified.

## 7.6 Joining ekpyrotic and Hagedorn phases, and a route to $\xi$

After the exclusion of constant  $w$ , the open question is what supplies  $w(t)$ . Joining two string-theoretic ingredients makes the requirements fit.

Flattening and smoothing during contraction operate only for  $w > 1$ , being a competition between curvature  $a^{-2}$ , shear  $a^{-6}$  and the background  $a^{-3(1+w)}$ . An ekpyrotic modulus with a steep negative potential supplies this [6]. A Hagedorn phase, by contrast, puts energy into oscillator and winding modes and has  $w \simeq 0$ , so it *cannot* flatten — the same point as the known observation that string gas cosmology does not by itself solve the flatness problem [7].

Joined *in sequence*, each phase covers the other’s deficit: the ekpyrotic phase flattens, and once the density reaches the Hagedorn scale that phase takes over and supplies the shallow equation of state at the bounce. Tested at  $w_f = 0$  (that is,  $p_f = 1.5$ ), the configuration passes every condition of Table 1:  $\min Q_s = 3.000$ ,  $\min Q_T = 0.500$ ,  $\min c_s^2 = 0.098$ ,  $\max c_s^2 = 0.820$ ,  $w(-\infty) = 4.33$ ,  $\Delta \ln(\sigma^2/\rho) = -6.55$ , with the coefficient gate passed at all 47 999 sampled points.

A possible by-product — a route to deriving  $\xi$  — was examined and *closed*. If the Hagedorn scale fixes when the transition occurs, then the EFT cutoff sits at the same place,  $\Lambda = M_s = \rho_c^{1/4}$ , and is no longer adjustable. Feeding that  $\Lambda$  into the gate gives

$$\frac{E_{\text{char}}}{\Lambda} = 0.519, \quad (12)$$

five times the 0.1 criterion, with 8 833 of 47 999 points violating EFT control. For constant  $w$  the analytic value is  $E_{\text{char}}/\Lambda = (4/3)^{1/4}/\sqrt{2} = 0.760$ , *independent of  $w$* . This is physically unsurprising: if the bounce occurs at the string scale, the characteristic energy of the bounce *is* the string scale, which is not a regime where an effective description is controlled.

Control would require  $\rho_H/\rho_c > 725$  — a bounce some seven hundred times below the string density. But then the Hagedorn scale no longer fixes  $\rho_c$ , and  $\xi$  is free again.

What survives and what does not. **Survives:** the stability of the joined background; ghost and gradient violations remain zero in both sectors even in the run above. **Dies:** the identification  $\rho_c = \rho_H$ , and with it the route to  $\xi$ . **Newly exposed:** if the bounce really is at the string scale, then the stability verdict of §7 was itself obtained with an effective description in a regime where that description is not controlled.

**Option (a): push the bounce below the string scale.** Accepting  $\rho_H/\rho_c > 725$  as a requirement, and returning  $\xi$  to being a free parameter, restores control. At  $\rho_H/\rho_c = 725$  one finds  $E_{\text{char}}/\Lambda = 0.1000$  (marginal) and at  $10^4$ , 0.0519 (comfortable), with zero EFT violations; stability is untouched in all cases.

The price is the curvature scale. With  $\rho_H/\rho_P \sim g_s^4$  one has  $\xi_{\text{red}} \equiv \rho_c/M_P^4 = g_s^4/N$ , and since  $H = 0$  at the bounce,  $\mathcal{K} = 12\dot{H}^2$ , so the coefficient  $\eta$  of equation (3) is  $\eta = 3\xi_{\text{red}}^2(1+w)^2$ .

Since  $\eta$  falls between  $10^{-16}$  and  $10^{-6}$ , **the curvature bound does not saturate at the Planck curvature** but six to sixteen orders of magnitude below it. The tacit reading of equation (2) with  $\eta \sim O(1)$  does not survive; the actual transition scale is  $\rho_c^{1/4} \sim 10^{16}\text{--}10^{17}$  GeV.

| $g_s$ | $N = \rho_H/\rho_c$ | $\xi_{\text{red}}$   | $\eta$                | $\rho_c^{1/4}$ [GeV] |
|-------|---------------------|----------------------|-----------------------|----------------------|
| 0.1   | 725                 | $1.4 \times 10^{-7}$ | $5.7 \times 10^{-14}$ | $4.7 \times 10^{16}$ |
| 0.3   | $10^4$              | $8.1 \times 10^{-7}$ | $2.0 \times 10^{-12}$ | $7.3 \times 10^{16}$ |
| 0.5   | 725                 | $8.6 \times 10^{-5}$ | $2.2 \times 10^{-8}$  | $2.4 \times 10^{17}$ |
| 1.0   | $10^4$              | $1.0 \times 10^{-4}$ | $3.0 \times 10^{-8}$  | $2.4 \times 10^{17}$ |

Table 2: Cost of restoring EFT control by placing the bounce below the string scale. The coefficient  $\eta$  of equation (3) lands far below unity.

That is the decade tensor-mode observations probe, though the bounce is not inflation and the bound on  $r$  does not transfer directly.

What is gained is EFT control, so the stability verdict is obtained where the description is trustworthy. What is lost is that  $\xi$  remains free and the Planck-scale reading of the bound does not hold.

Three caveats remain. (i)  $w_f = 0$  approximates the Hagedorn value; the string-gas dynamics was not computed. (ii) No equation drives  $w$  from 4.33 to 0 — the test inserts a tanh profile by hand. (iii) The cutoff remains an assumption, and it is now known that the assumption is not harmless.

### 7.7 What this section does not claim

1. The effective-field-theory cutoff is not derived.
2. No microscopic mechanism fixes the relic amplitude; only its shape is predicted.
3. The entropy field and its 0.29% tuning are inserted by hand.
4. The phase joining above supplies a background that passes the stability conditions; it does not supply the dynamics driving the transition, and it fails EFT control if the cutoff is identified with the string scale.
5. No data analysis was carried out. The analysis design is pre-registered, but no CMB map was opened.

## 8 Limitations and next steps

- **Open.** A microscopic derivation of  $\eta$ ,  $\xi$  and the transition dynamics. The EFT cutoff  $\Lambda$  belongs here: it is the one entry of the stability gate that remains an assumption.
- **Partly settled.** Stability including anisotropies. Scalar and tensor second-order perturbations are decided over the whole range; anisotropy is computed only at the level of a perturbative test shear.
- **Open.** Global matching from a black-hole interior to an expanding region.
- **Open.** Information, entropy and Hawking radiation.
- **Partly settled.** Observables comparable with data. The scalar tilt and the quadrupole shape are computed; the amplitude is not, and the gravitational-wave side is absent.

## Conclusion

A finite curvature or density scale remains a useful research hypothesis for replacing a classical singularity with a non-singular transition. The verification reported here moved the proposal in two directions at once. It advanced, in that the effective background belongs to a family with a stable microscopic realisation, and that a primordial spectrum and a quadrupole shape follow from it by calculation. It narrowed, in that the same calculation excluded the minimal single-field realisation, the constant-equation-of-state baseline, and the standing of the bounce’s adiabatic mode as the origin of the CMB.

What survives is a single quadrupolar signal whose shape is predicted and whose size is not, resting on an anisotropy put in by hand. The value of this work lies not in declaring an answer to quantum gravity, but in recording which of the required items turned out to be computable — and which did not survive being computed.

## Data and code availability

All numerical results were produced by scripts that read no observational data. Analysis code, gate criteria, simulation outputs and the pre-registered CMB analysis design are maintained with the manuscript and are available from the author on request.

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